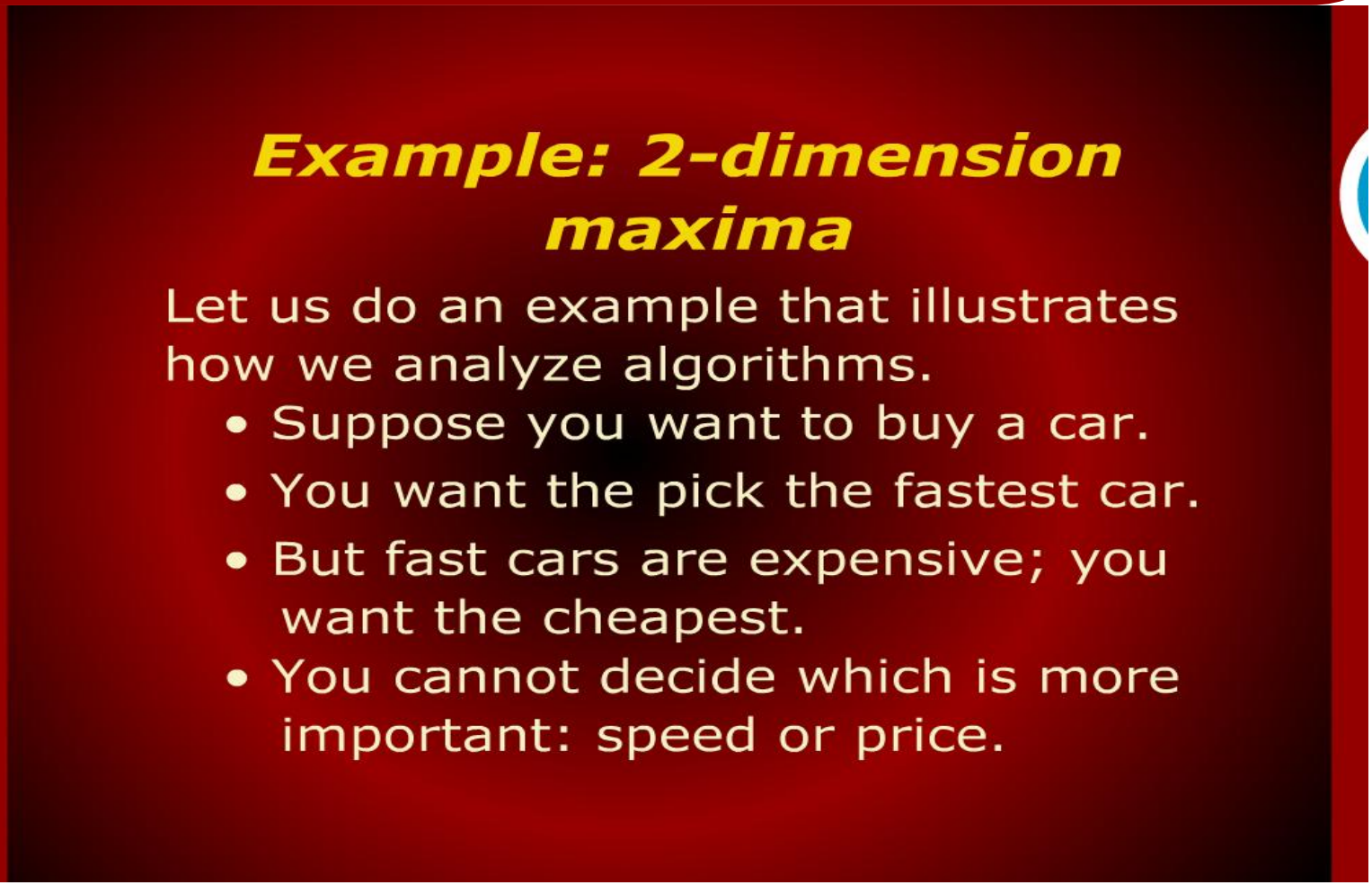
Algorithms

Lecture # 02

Syed Hamed Raza

Example: 2-dimension maxima



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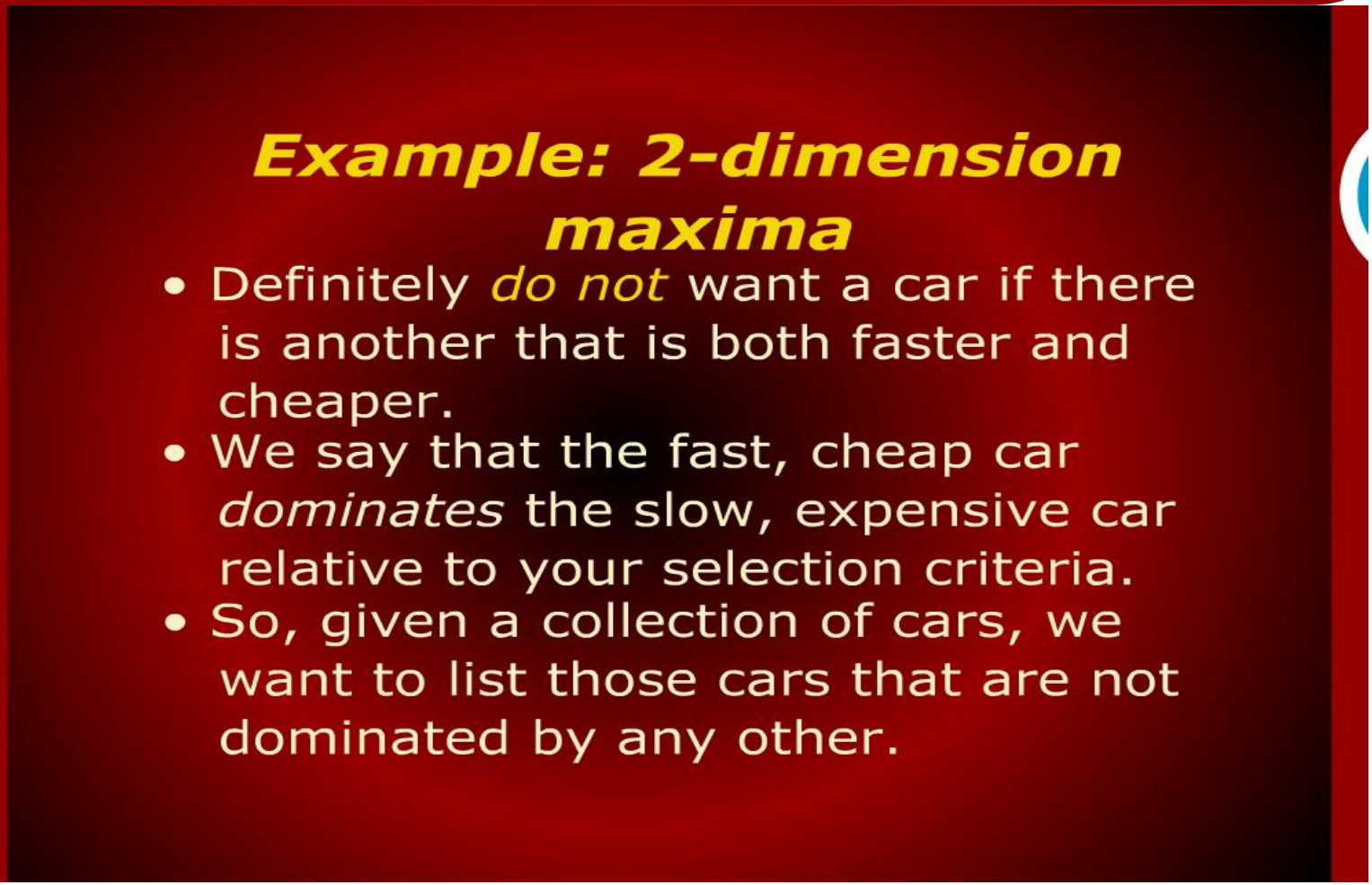

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Here is how we might model this as a formal problem.

- Let a point p in 2-dimensional space be given by its integer coordinates, $p = (p.x, p.y)$.
- A point p is said to be *dominated* by point q if $p.x \leq q.x$ and $p.y \leq q.y$.

Example: 2-dimension maxima

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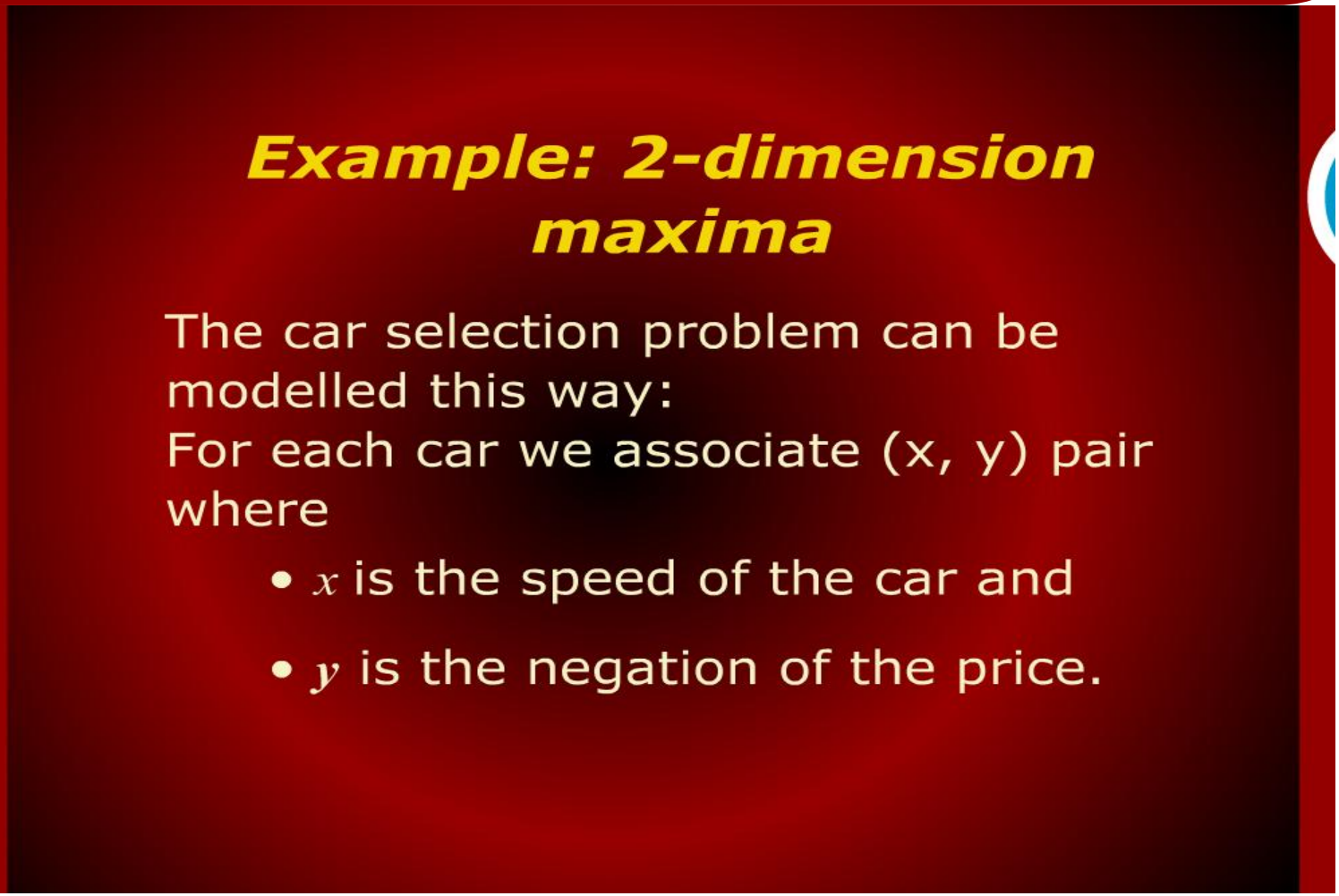
- Given a set of n points, P = $\{p1, p2, \ldots, pn\}$ in 2-space a point is said to be *maximal* if it is not dominated by any other point in P.

Example: 2-dimension maxima



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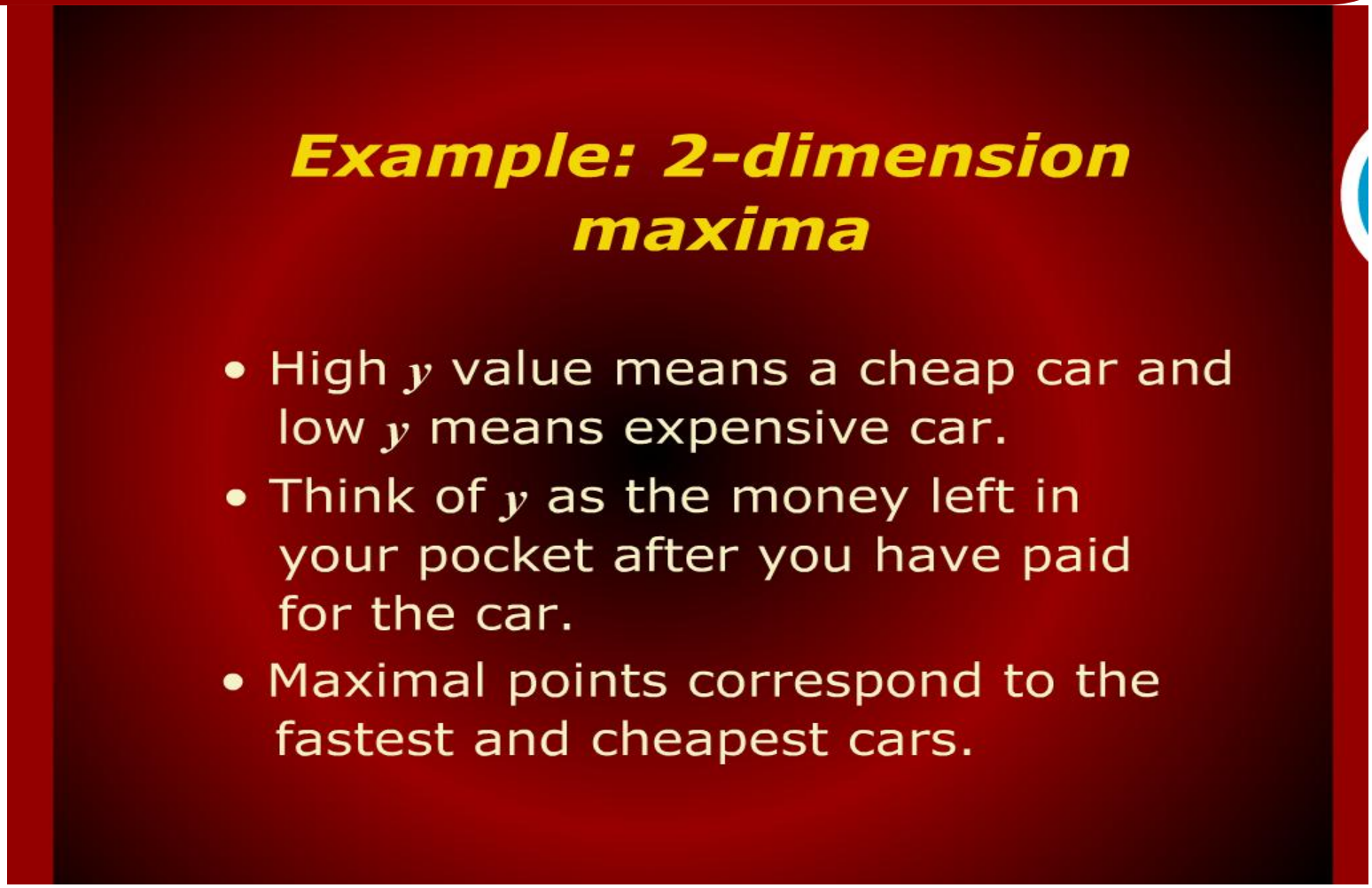

Example: 2-dimension maxima

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2-dimensional Maxima:

- Given a set of points $P = \{p_1, p_2, \ldots, p_n\}$ in 2-space,
- output the set of maximal points of P,
- i.e.,those points p_i such that p_i is not dominated by any other point of P.

Example: 2-dimension maxima

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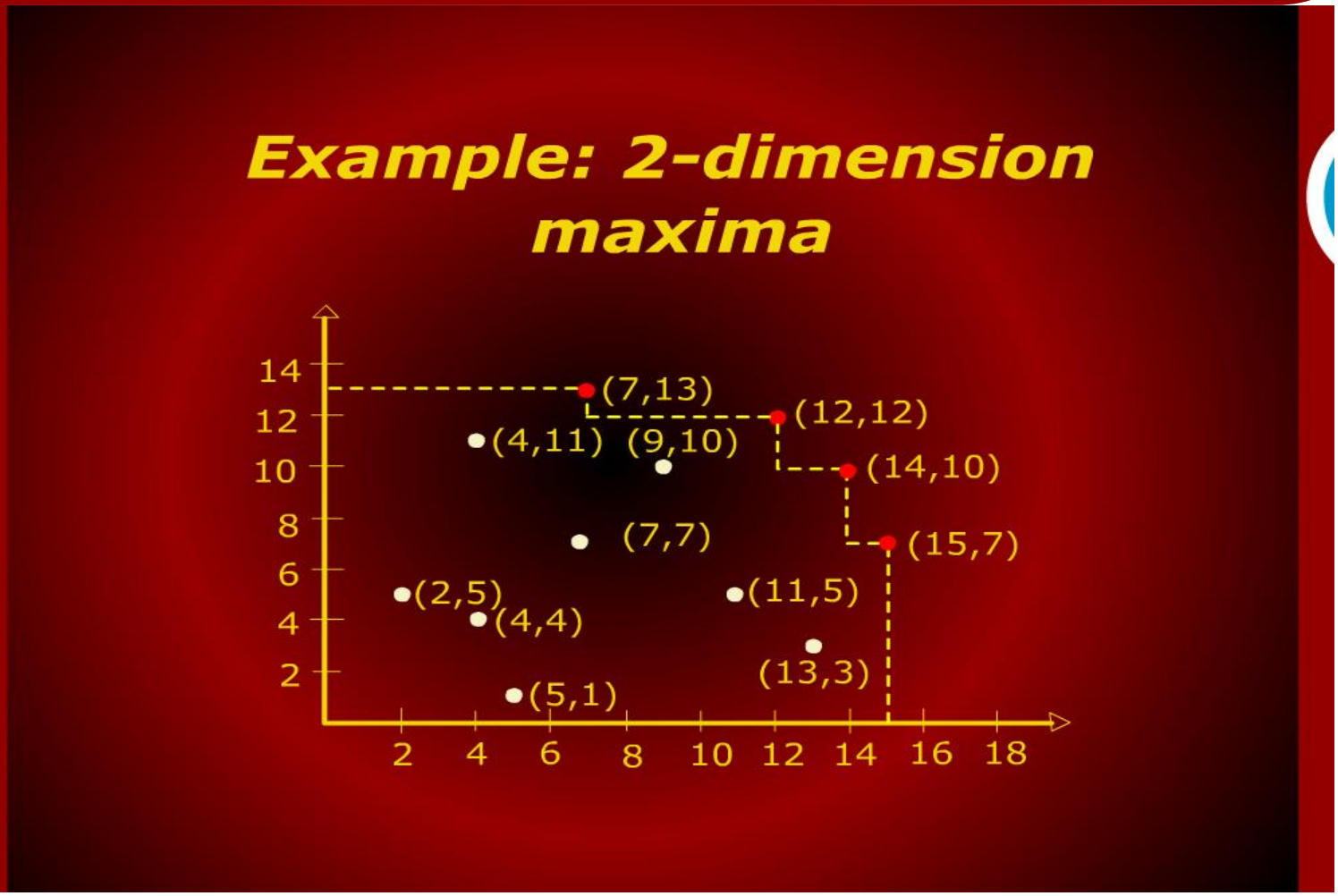
- Our description of the problem is at a fairly mathematical level.
- We have intentionally not discussed how the points are represented.
- We have not discussed any input or output formats.
- We have avoided programming and other software issues.

Example: 2-dimension maxima



Example: 2-dimension maxima

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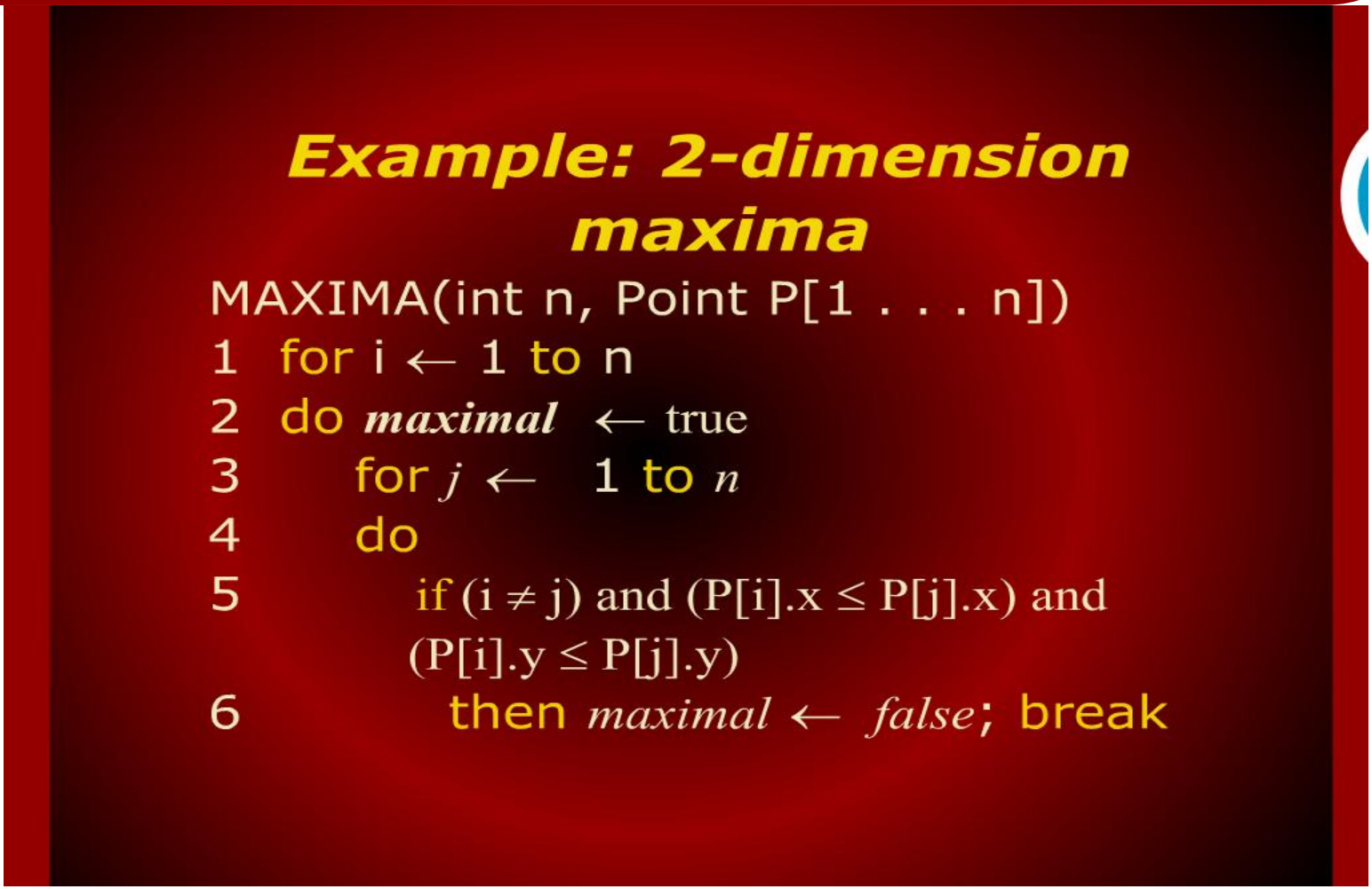
Brute-Force Algorithm:

- Let $P = \{p_1, p_2, \ldots, p_n\}$ be the initial set of points.
- For each point p_i, test it against all other points p_j.
- If p_i is not dominated by any other point, then output it.

Example: 2-dimension maxima



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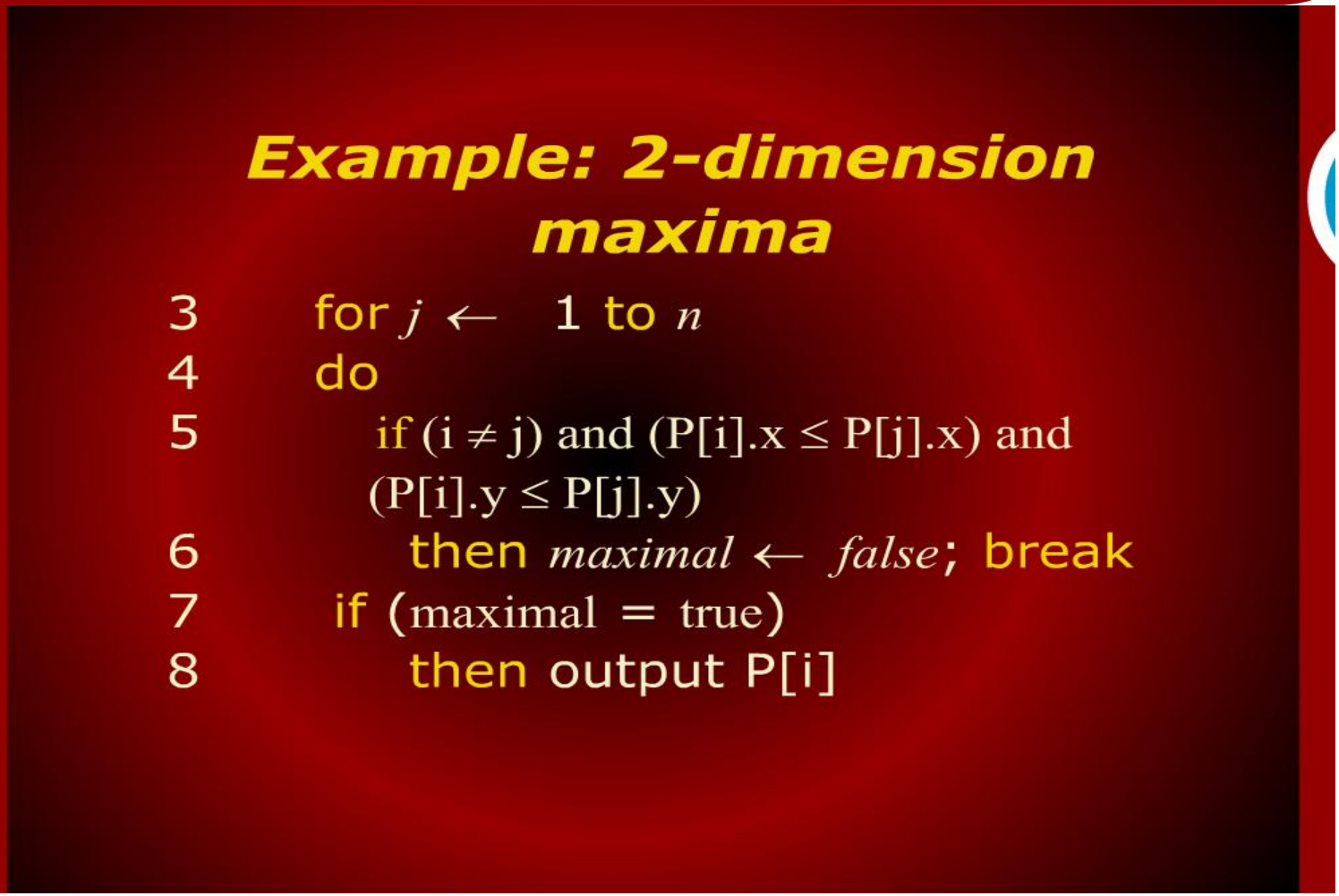

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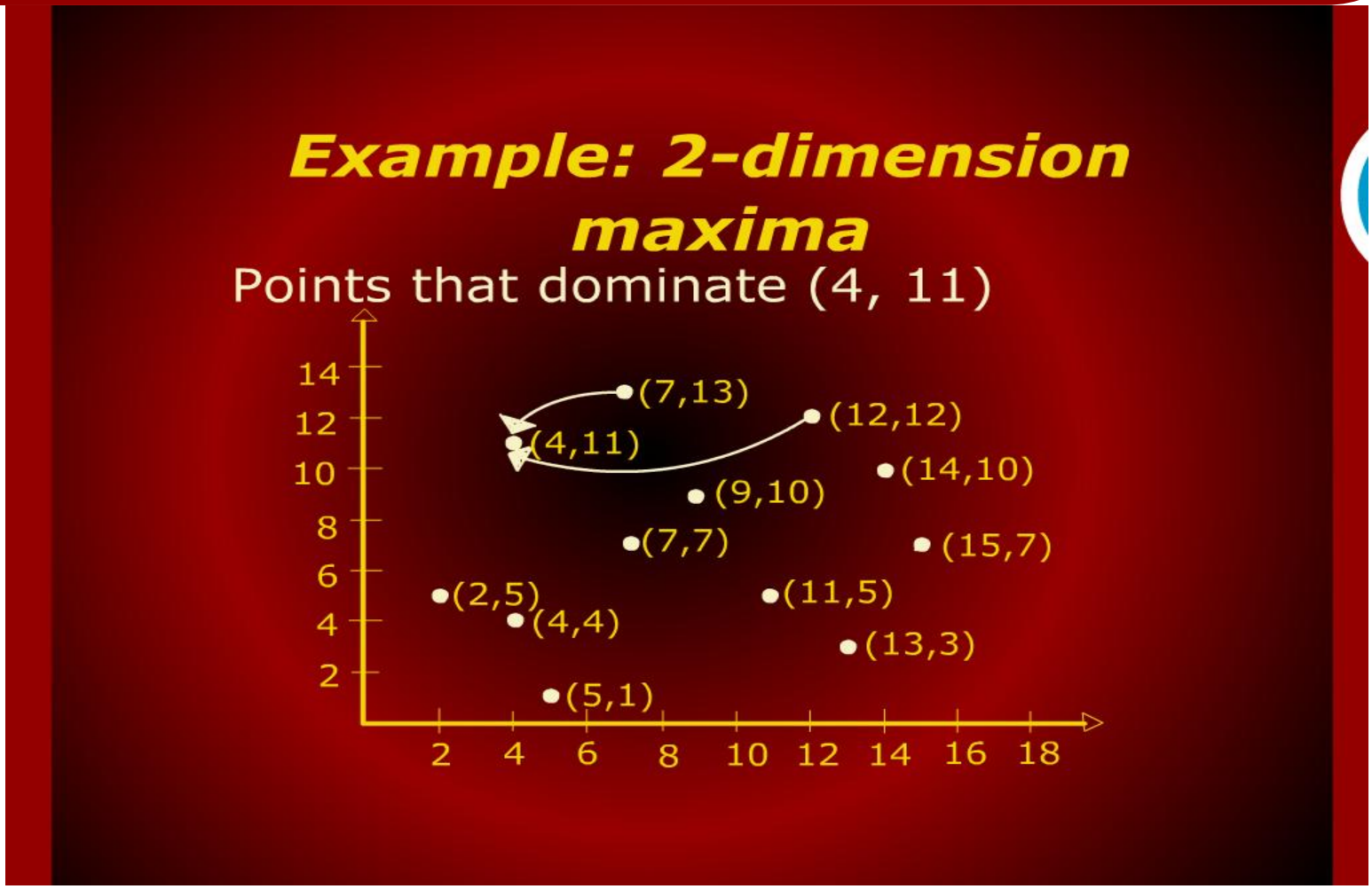

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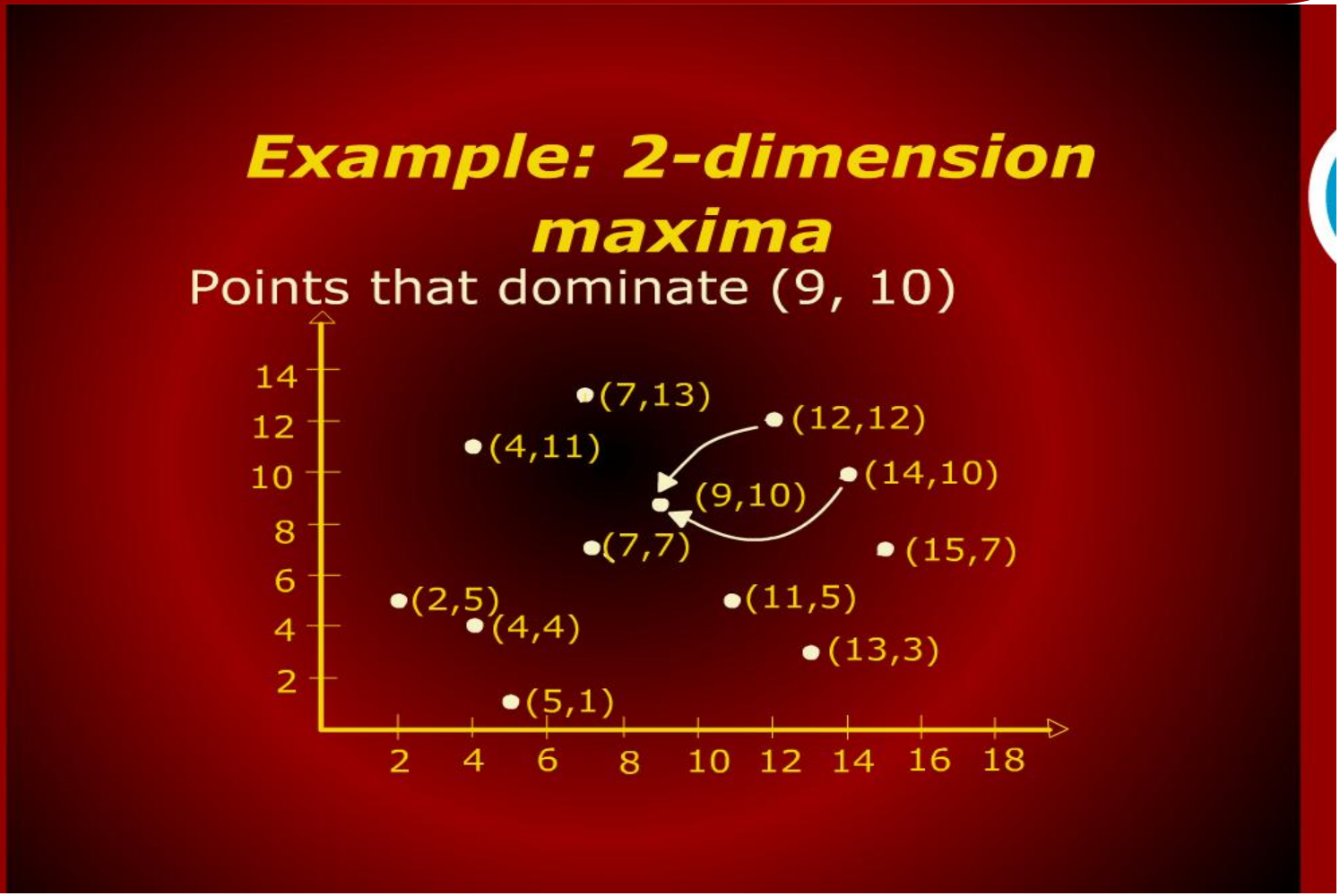

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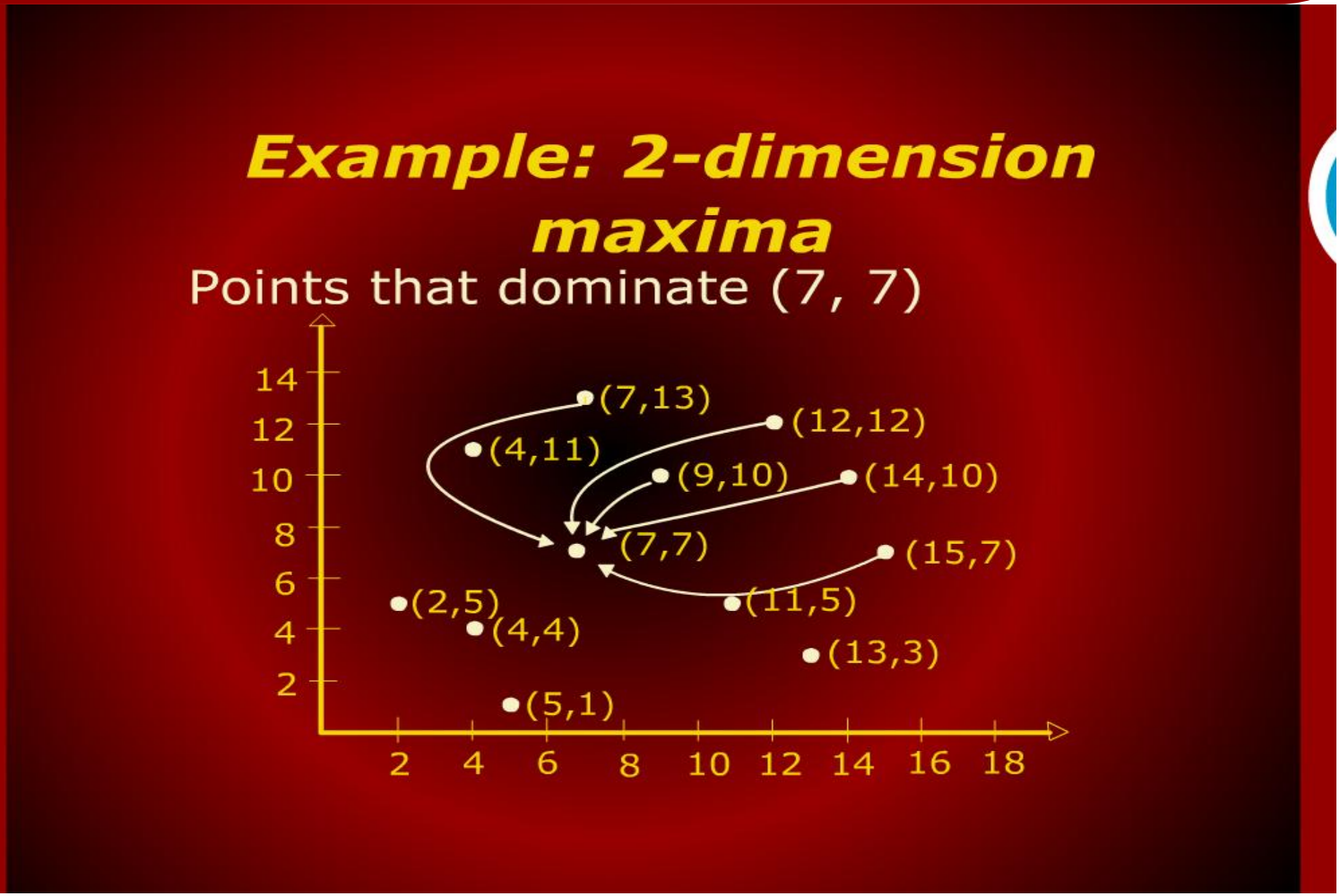

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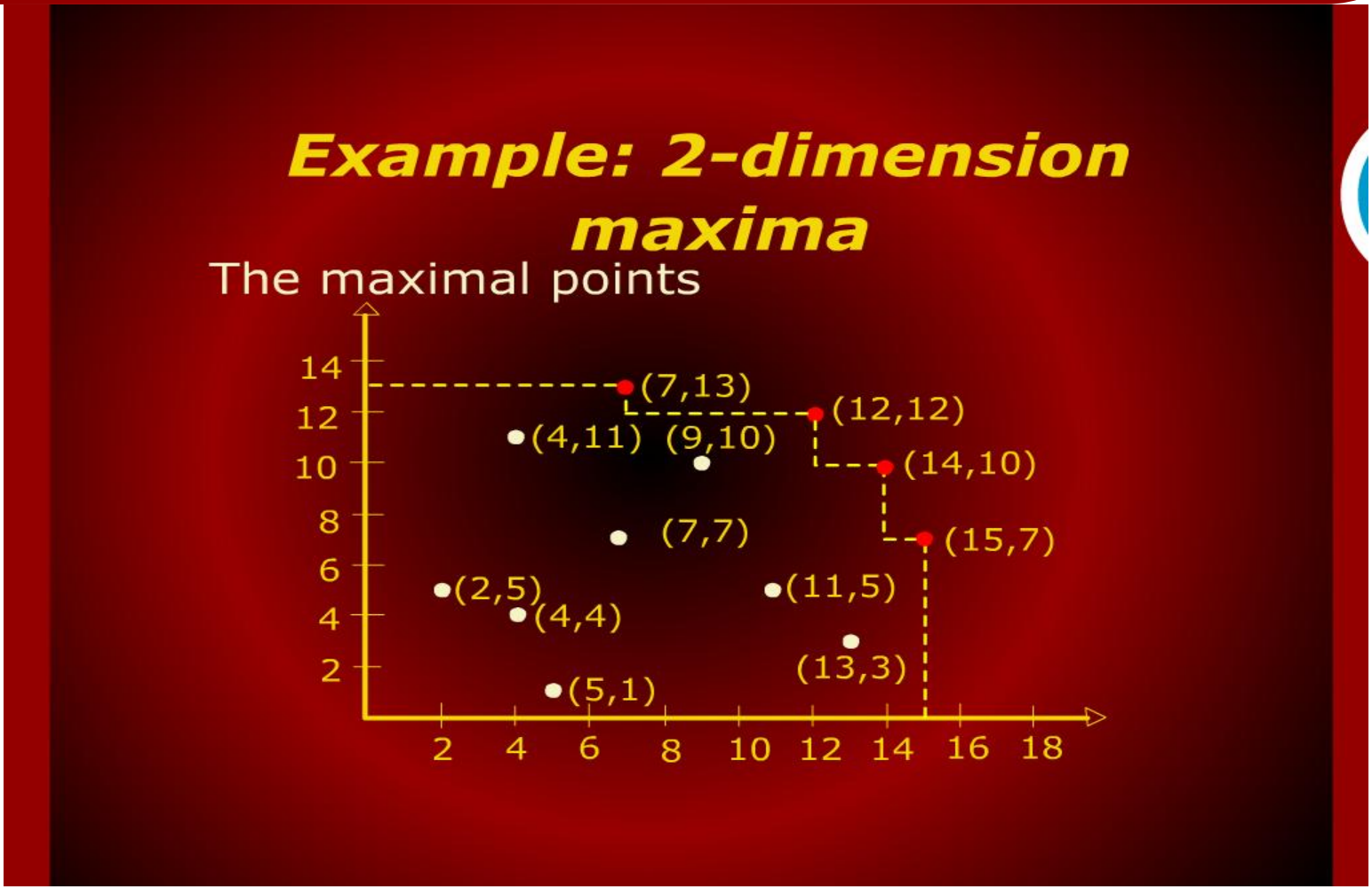
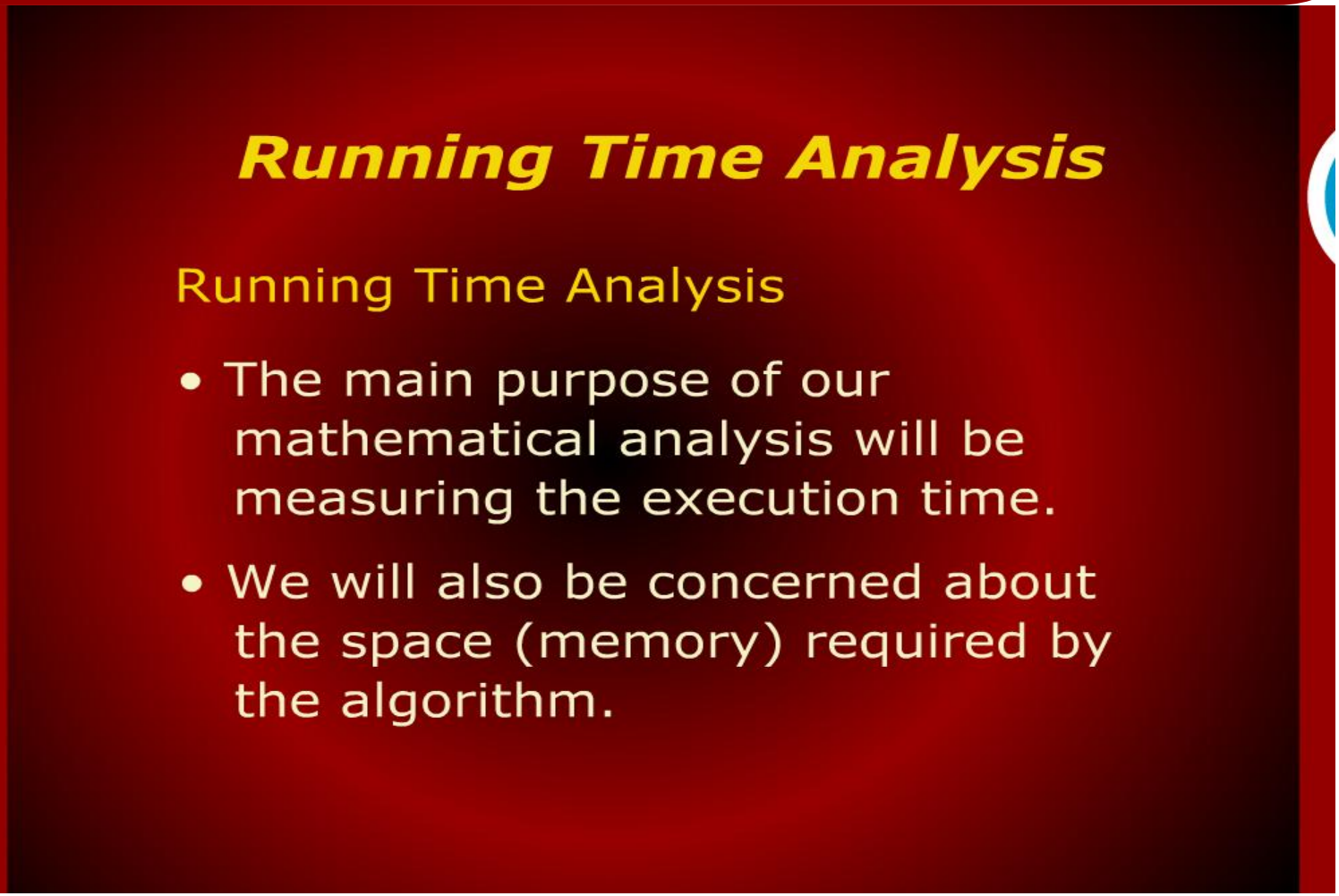
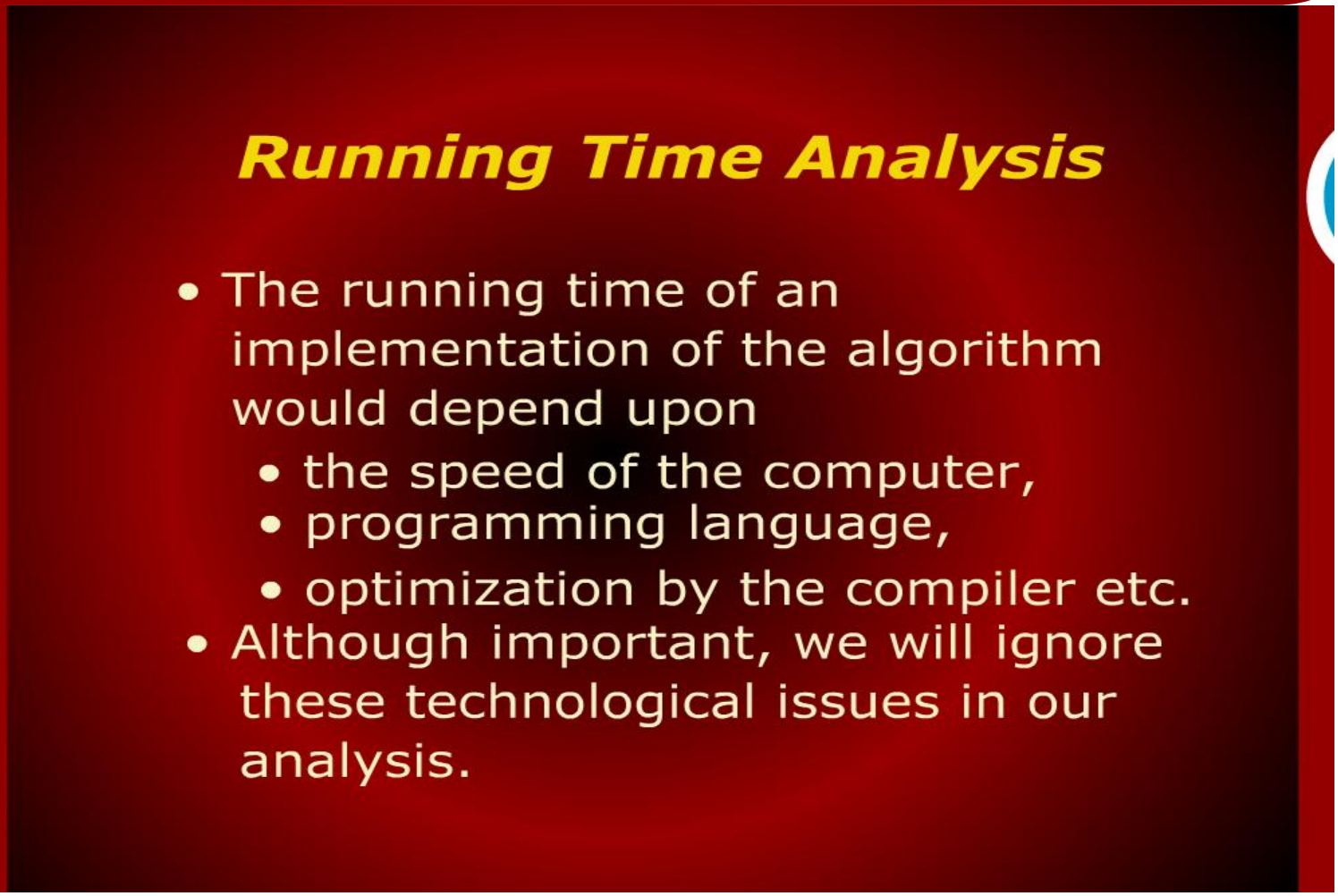


Running Time Analysis



Running Time Analysis



Analysis of 2-d Maixma

Analysis of 2-d Maxima

To measure the running time of the brute-force 2-d maxima algorithm,we could

- count the number of steps of the pseudo code that are executed
- or, count the number of times an element of P is accessed
- or, the number of comparisons that are performed.